

**Project Title:- Mind Guessing System
(MindSpark)**

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1. Introduction

The **Mind Guessing System** is an intelligent interactive tool that predicts what the user is thinking by asking yes/no/maybe questions. It uses attribute-based filtering and a structured database of animals, characters, and objects to narrow down possibilities. By selecting high-priority attributes, the system quickly eliminates options and makes an accurate final guess.

Objectives

- To predict the user's thought using attribute-based filtering and systematic questioning.
- To use priority-based attributes for faster and more accurate elimination of possibilities.
- To maintain a structured and expandable database of animals, characters, and objects.
- To create an engaging interactive system that improves accuracy through user responses.

2. Problem Statement

Simple Description of the problem

- People often think of an animal, character, or object and expect the system to guess it accurately, but traditional guessing methods rely on random questions, making the process slow and inaccurate. Identifying the correct item becomes difficult without structured logic, leading to confusion and repeated questioning. There is a need for a user-friendly tool that can intelligently narrow down possibilities and guess the user's thought efficiently using a systematic approach.
- The Mind Guessing System addresses this gap by providing an automated, attribute-based prediction method that analyzes user responses, filters options logically, and delivers accurate guesses within a limited number of questions.

3. Basic Concepts

Key AI Terminologies

- **Attribute-Based Reasoning:** A logic-driven approach where each item is defined by unique attributes, allowing the system to filter and match options based on user responses.
- **Knowledge Base:** A structured dataset containing animals, objects, and characters along with their attributes, used for comparison during prediction.
- **Priority-Based Questioning:** A method where highly significant attributes are asked first to speed up filtering and increase the accuracy of the final guess.
- **Logical Filtering:** A process where items are included or removed from the candidate list depending on whether user responses match their attributes.

Introduction to AI Techniques Used in the Project

- **Rule-Driven Knowledge Base Reasoning:** The system applies predefined rules and attribute sets to match user responses with stored data, ensuring structured and logical prediction.
- **User-Guided Learning:** If the system predicts incorrectly, the user's input is saved to improve and expand the knowledge base for future interactions.
- **Attribute-Based Filtering Algorithm:** User replies (yes/no/maybe) are used to filter items step-by-step, allowing quick elimination and accurate narrowing of possible results.
- **Intelligent Image Retrieval Using Google Custom Search API:** The system retrieves real-time images of the final predicted item through Google's Custom Search API, providing visual confirmation for the user.

4. Requirement Analysis

Tools and Technologies Needed

✦ Frontend:

- **HTML5:** Provides the structure for all pages, including category selection, question display, and result screens.
- **CSS3:** Used to style the interface, ensuring a clean, simple, and user-friendly design.

✦ Backend:

- **Python (Flask Framework):** Handles game logic, routing, user sessions, question flow, attribute filtering, and final prediction generation.
- **JSON Files:** Serve as the knowledge base that stores animals, objects, and characters with their corresponding attributes.

✦ Additional Tool:

- **Google Custom Search API:** Retrieves real-time images of the predicted item, improving user experience with visual confirmation.

Basic System Requirements

✦ Hardware:

- **Processor:** Minimum dual-core (i5 or equivalent recommended)
- **RAM:** At least 4GB (8GB recommended)
- **Storage:** Around 5–10GB free space for project files and dependencies

✦ Software:

- **Python (Version 3.8 or higher)** for running the Flask backend.
- **Flask Framework** installed via pip.
- **Code Editor** such as VS Code or any Python-supported editor

✦ Dependencies:

- **Python Libraries:** Flask, Requests
- **JSON Files:** Used as the knowledge base
- **Google Custom Search API:** For fetching final guess images

5. Dataset

Source of data

The dataset used in the **MindSpark Guessing System** is **fully self-created** and **manually curated**.

No external APIs, no public datasets, and no pre-trained knowledge bases were used.

❖ Data was collected through:

- **Manual Research:**

- Listing commonly known **animals, objects, and fictional characters**.
- Assigning **meaningful and unique attributes** to each item based on general knowledge and observation.

- **Multiple Information References:**

General reference sources were used only for conceptual understanding, such as:

- Wikipedia articles
- Popular knowledge websites
- Books and real-life observations

Overview of the data

The dataset used in the Mind Guessing System consists of **three structured JSON files** that store predefined items and their attributes. These datasets form the core knowledge base of the system, enabling logical filtering and prediction.

- **Animals Dataset (animals.json)** : Contains a list of animals along with unique identifying attributes such as habitat, characteristics, behaviours, and physical traits.
- **Objects Dataset (objects.json)** : Includes various household, daily-use, and general objects, each defined through specific functional and physical attributes.

- **Characters Dataset (characters.json)** : Stores real-life personalities, fictional characters, and animated figures with attributes describing their role, appearance, background, and identity.

Basic Data Cleaning

Since the dataset is manually created, several cleaning steps were required to ensure accuracy and consistency:

1. Duplicate Removal:

- Removed repeated items across categories.
- Ensured no item had identical or redundant attributes.
- Verified that each entry was unique in name and meaning.

2. Attribute Standardization:

All attributes were converted to **lowercase** and **snake_case** format.

Example:

“Fast Runner” → “fast_runner”

“Black & White Stripes” → “black_white_stripes”

Removed spaces, special characters, and inconsistent naming styles.

3. Uniform Structure:

Every item in the dataset follows the same clean format:

```
Json
{
  "name": "Tiger",
  "attrs": ["carnivore", "orange_black", "fast_runner", "wild"]
}
```

This uniformity helps the filtering algorithm and question generation module work smoothly.

4. Validation Checks:

- Removed overly generic or meaningless attributes like “common”, “normal”.

- Ensured each animal/object/character has **at least one unique identifying attribute**.
- Validated that attributes are relevant and factually correct.

5. Category Separation:

For better clarity and accuracy, the dataset is divided into three separate JSON files:

- **animals.json**
- **objects.json**
- **characters.json**

This prevents confusion during:

- attribute filtering
- question selection
- category-based guessing

6. Proposed Solution

Simple Explanation of the Approach

- The proposed solution is an interactive web-based system where the user thinks of an animal, object, or character, and the system tries to guess it by asking a series of yes/no/maybe questions. Each item in the database is defined using unique attributes stored in JSON files.
- Based on each yes/no/maybe answer, the system removes items that do not match, steadily reducing the possible candidates.
- It chooses the next question using priority-based attributes, helping eliminate options faster.
- When only a few candidates remain, the system makes its final guess and displays a real-time image using the Google Custom Search API.
- If the guess is wrong, the user can provide the correct name and attributes, which are saved to improve the knowledge base.

Algorithm or Model Chosen

In this mind-guessing system, four primary techniques are used — Attribute-Based Filtering, Priority-Based Question Selection, Rule-Driven Reasoning, and Google Image Retrieval.

- **Attribute-Based Filtering Algorithm:** The system compares each user answer (yes/no/maybe) with stored attributes and continuously filters out items that don't match. This step-by-step elimination helps narrow down the correct guess efficiently.
- **Priority-Based Question Selection:** Questions are chosen based on predefined attribute priority lists, ensuring high-impact attributes are asked first to reduce the search space quickly.
- **Rule-Based Reasoning Engine:** Instead of using machine learning, the system follows logical rules to evaluate responses, score matches, and identify the closest remaining item.

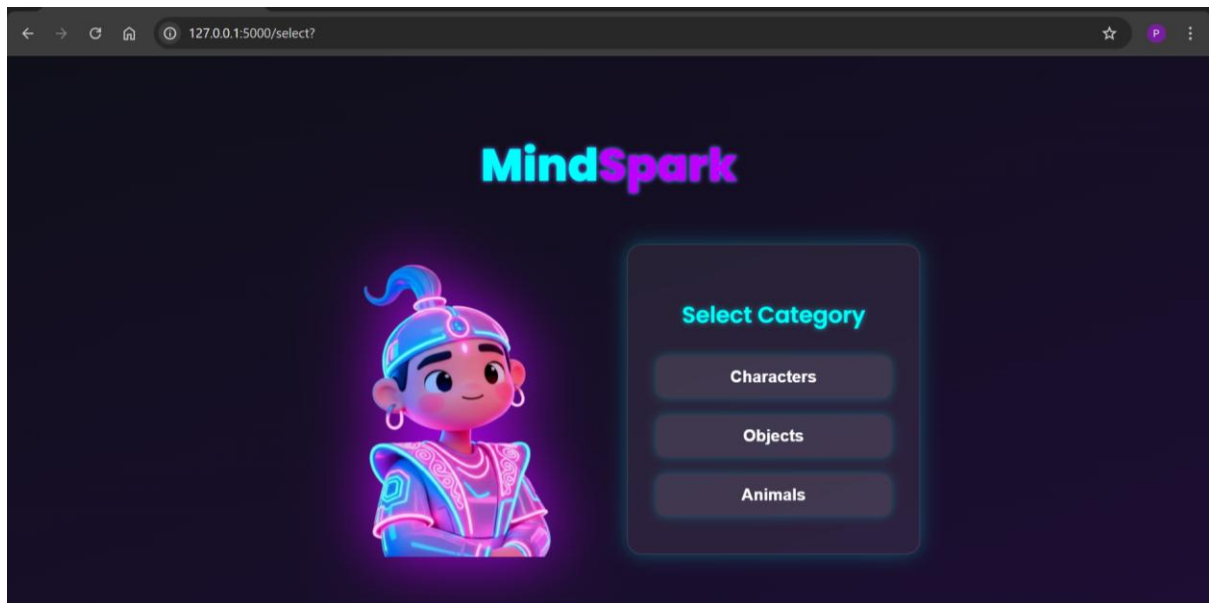
- **Google Image Retrieval Module:** Once the final guess is made, the system fetches a real-time image using the Google Custom Search API for visual confirmation.

7. Implementation

+ Step by step process

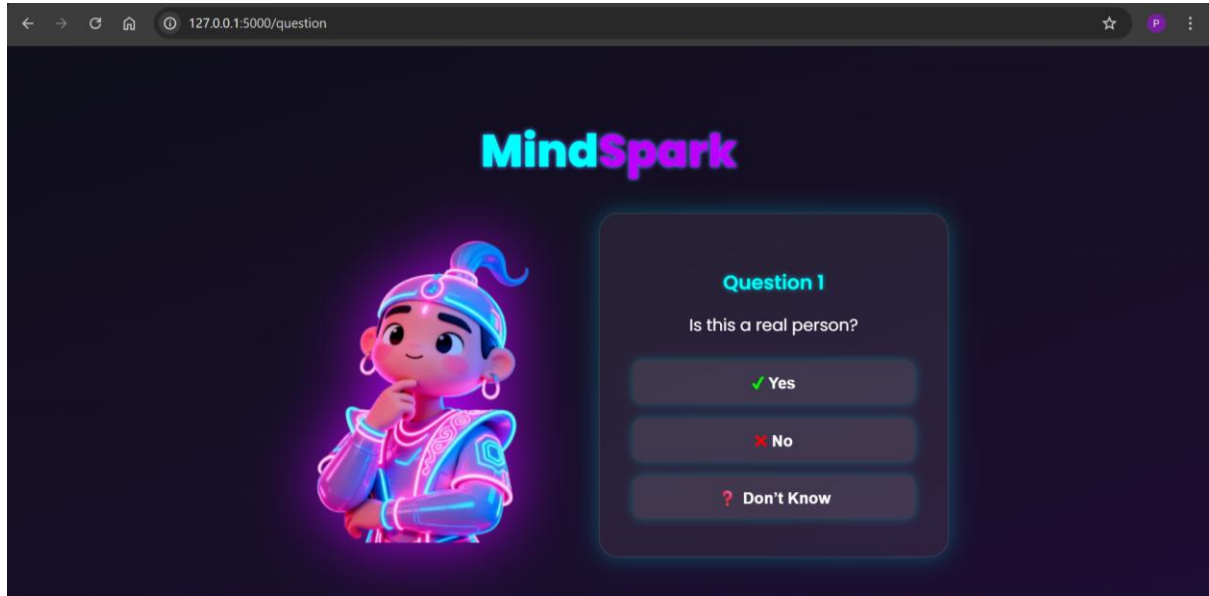
1. Category Selection

- The user begins by selecting a category — Animal, Object, or Character.
- This selection decides which JSON knowledge base will be used.



2. Question Generation

- Based on the selected category, the system loads all items and their attributes.
- It then chooses the most important attribute (using priority lists) and generates a question.



3. User Response Processing

- When the user answers yes / no / maybe, the system compares the response with item attributes and:
- Keeps items that match the answer
- Removes items that contradict it
- This gradually reduces the list of possible candidates.

4. Attribute-Based Filtering

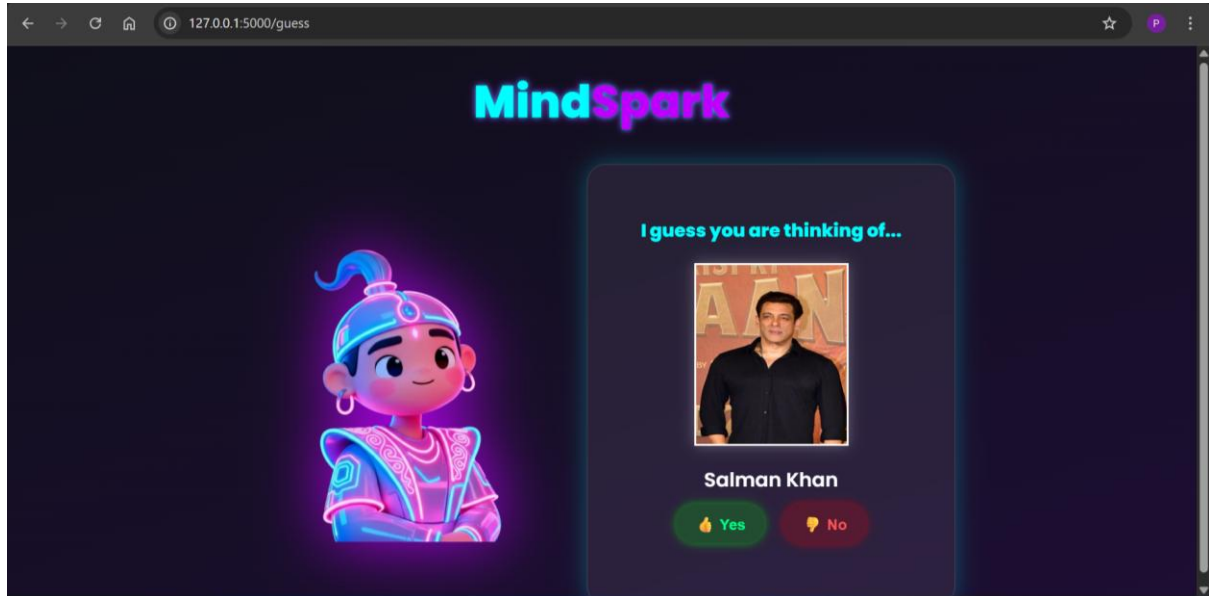
- After each answer, the algorithm repeatedly filters the knowledge base to remove mismatched items, ensuring the remaining list becomes smaller and more accurate.

5. Priority-Based Question Selection

- The system selects the next most informative attribute using predefined priority lists so that unnecessary questions are avoided and the correct guess appears faster.

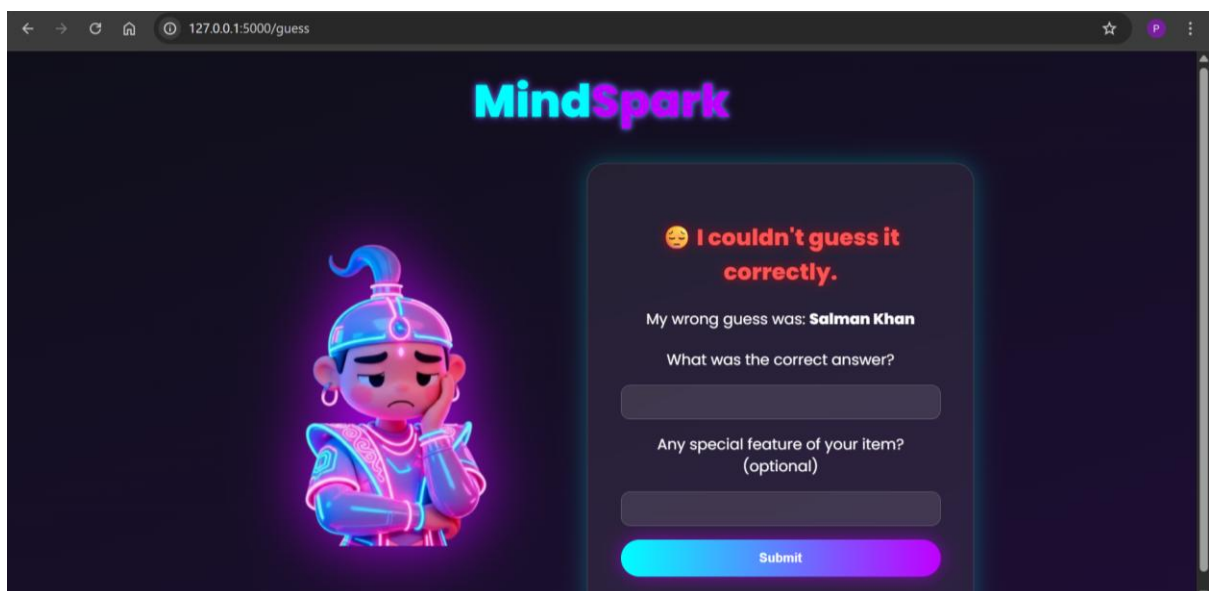
6. Final Prediction

- When the remaining candidates become very few (sometimes just one), the system makes a final guess and displays:
- The predicted name
- A real-time image fetched using the Google Custom Search API



7. Learning From User Feedback

- If the guess is wrong:
- The user can enter the correct name
- Additional details can be provided
- The system saves the entry into a pending JSON file, improving the knowledge base for future gameplay



This approach combines simple rules, structured attributes, and priority reasoning to create a fast, efficient, and intelligent guessing system.

8. Testing

Simple Test Cases

Test Case	Input	Expected Output	Actual Output	Status
1. Basic Object Guessing	User answers attributes for a known object (e.g., mobile phone)	System filters items and correctly guesses the object	Matches Expected Output	Pass
2. Incorrect Answers Flow	User purposely gives wrong answers for some attributes	System adapts, removes mismatched items, and asks new attributes	Matches Expected Output	Pass
3. Ambiguous Object	Multiple objects share many attributes	System continues asking priority attributes until only one remains	Matches Expected Output	Pass
4. User Says “Don’t Know”	User answers “don’t know” for an attribute	System skips filtering for that attribute and asks the next best one	Matches Expected Output	Pass
5. Very Unique Object	Object contains rare or unique attributes	System quickly narrows down based on priority attributes	Matches Expected Output	Pass
6. Category Mismatch	User answers cause all candidates to be removed	System shows: “Object not found. Try again.”	Matches Expected Output	Pass
7. Repeated Attribute Prevention	System might accidentally select same attribute again	System checks asked list and never repeats the same question	Matches Expected Output	Pass

9. Challenges

Basic Problems Encountered

1. **Attribute Overlap & Ambiguity:** Many animals, objects, and characters shared similar attributes, causing confusion during filtering and making it difficult for the system to narrow down the correct item.
2. **Question Selection Issues:** The priority list did not always pick the most informative attribute first, resulting in slow elimination and extended questioning.
3. **Incorrect or Uncertain User Responses:** Answers such as “don’t know”, or incorrect yes/no inputs due to confusion, weakened the filtering process and sometimes eliminated correct candidates.
4. **Knowledge Base Limitations:** The JSON database initially contained a limited number of items, leading to frequent “not found” results for unique animals, objects, or characters that were not present in the system.
5. **Final Guess Conflicts:** When multiple candidates shared nearly identical attribute patterns, the system had difficulty selecting the correct final guess, reducing accuracy.

How They Were Resolved

1. **Improved Attribute Structuring:** Attributes were reorganized to reduce overlap and increase distinctiveness. Clear differentiating features were added so the system could more accurately separate similar items during filtering.
2. **Enhanced Priority-Based Question Selection:** The priority lists for animals, characters, and objects were refined to ensure that highly informative attributes are asked first. This significantly improved elimination speed and reduced unnecessary questions.
3. **Better Handling of “Don’t Know” and Incorrect Responses:** A neutral logic was introduced for “*don’t know*” responses, preventing incorrect filtering. Additional safety rules were added to avoid removing valid candidates due to occasional user mistakes in yes/no inputs.

4. **Improved Final Prediction Logic:** A weighted scoring and tie-breaker mechanism was implemented to handle cases where multiple candidates were similar. This strengthened the accuracy of the final guess and reduced incorrect predictions.
5. **Expanded & Learnable Knowledge Base:** A pending_updates.json system was introduced, enabling new entries to be added through user feedback without affecting the core database. This allowed continuous growth and improved coverage.

This refined approach solved major technical and UX issues, improving accuracy and making the system more user-friendly.

10. Conclusion

What Was Learned?

- 1. Efficiency of Attribute-Based Reasoning:** Using structured attributes and logical filtering proved to be an effective method for predicting the user's thought without requiring complex AI or machine learning models.
- 2. User Interaction Importance:** Clear questions, smooth flow, and easy-to-understand responses significantly enhanced user engagement and satisfaction during the guessing process.
- 3. Value of Systematic Questioning:** Priority-based question selection and organized attribute mapping showed how a structured approach can drastically improve accuracy and reduce unnecessary questions.
- 4. Need for Robust Data Structure:** Challenges like overlapping attributes and limited initial data highlighted the importance of a clean, well-organized, and continuously improvable knowledge base.

Key Outcomes of the Project

- **Functional Mind Guessing System:** Successfully developed a system capable of narrowing down user-selected animals, characters, or objects through logical questioning and delivering accurate final predictions.
- **User-Friendly Gameplay Experience:** Implemented features such as “don't know” handling, non-repetitive questions, final guess confirmation, and real-time image retrieval for an engaging and interactive experience.
- **Strong Foundation for Future Expansion:** While this version uses rule-based reasoning and attribute filtering, it creates a solid base for future integration of advanced AI models, automated knowledge updates, and larger datasets.

Future Scope

- **Score-Based Intelligence Improvement:** A scoring mechanism can be introduced where each attribute match increases the confidence score of an item. The system can use these scores to make more accurate predictions, reduce ambiguity, and even display confidence percentages in the final guess.
- **Expansion of Knowledge Base:** More categories and items (e.g., places, movies, gadgets, professions) can be added to make the system more versatile. With a larger and richer dataset, the model will handle a wider variety of user choices.
- **Adaptive Learning Mode:** The system can automatically learn from user corrections. Whenever the guess is wrong and the user provides the right item, the system can intelligently analyze new attributes and update its knowledge base without manual editing.

11. References

- **Flask Documentation** – <https://flask.palletsprojects.com/>
- **Python JSON Module** – <https://docs.python.org/3/library/json.html>
- **Google Custom Search API Documentation** – <https://developers.google.com/custom-search/v1/overview>

12. Appendix

 **GitHub Link:-**

<https://github.com/pooja877/Mind-Guessing-System.git>